

Search and Rescue– Unknown Victim Location Scenario

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I. PROJECT OVERVIEW

The design of the PDDL/+ model primarily focuses on the logic used for search and rescue robots after natural or man-made disasters. The comprehensive design adheres to an explore-detect-rescue architecture. Exploration relies on a topological map; detection is demonstrated using sensory technologies, and rescue operations transmit the information collected by the robot's exploration to search and rescue (SAR) teams. The design choices for each question enhance the applicability of PDDL/+ and the constraints associated with each model.

II. DESIGN LOGIC:

The logical design of this PDDL/+ is derived from the technology incorporated into robots for search and rescue (SAR) and the protocols that SAR teams must adhere to during natural or man-made disasters resulting in collapsed structures. This system mimics the exploration and assessment of victims with chemical sensors, thermal imaging, camera detection, and microphones. Chemical sensors and thermal imaging are mostly employed to assess victims' vital signs using specific methodologies that will be elaborated further. The camera detection, thermal imaging, and microphones facilitate the robot's autonomous navigation, with microphones aiding in locating the victim, while the camera and thermal imaging verify the presence of the victim.

Monitoring battery life, victims' health, and rescue duration correlates with the success rate of the rescue operation. If the battery life is insufficient for the duration of the rescue, it is a significant waste of time for SAR teams. In the event of a building collapse or approaching natural disaster, it is imperative to monitor rescue times to guarantee the safety of SAR personnel and victims. If SAR anticipates that the victim's condition will worsen after evaluation, they must regrettably redirect their resources to maintain search efficiency.

The rescue operations are designed to signal to SAR teams the path taken by the robot. This assists SAR personnel in mapping the victim's location within the structure's layout and, therefore, identifying the most efficient entry route to minimise additional structural damage.

Chemical sensing technology:

This technique was deployed in a robot known as SMURF. It enables the robot to distinguish between living and deceased victims using a "fuzzy logic algorithm". The research encompassed a series of criteria for both living and deceased victims. The technique is needed to identify volatile organic compounds (VOCs) found in the faeces, urine, breath, skin, milk, blood, and saliva of living victims. Conversely, for deceased individuals, it was necessary to identify specific volatile organic compounds (VOCs), including cadaverine, putrescine, skatole, and indole. The highest probability of accurately identifying either deceased or living victims was attained when the fuzzy logic system was used and the robot was situated 2-3 metres from the victim. [b4]

Thermal Cameras:

Thermal imaging aids in identifying the presence of living beings. Thermal cameras can be used to conduct a more thorough analysis of the situation and assist in establishing the victims' vital signs. The thermal camera technology, in conjunction with a series of particular algorithms, executes this operation. This combination facilitates the assessment of skin surface temperature, which may signify fevers; identifies breathing rate by monitoring temperature variations around the nose region; and detects blood circulation, utilised for determining the heart rate. This information assists SAR personnel in effectively addressing the scenario. [b3],[b5]

Video Camera:

The preliminary execution and discussion of this section are taken from a study that concentrated on the comparison between SAR operators and machine learning algorithms in recognising humans in aerial images.

The application of computer vision techniques, in conjunction with machine learning algorithms, aids search-and-rescue (SAR) teams in evaluating photographs to more effectively locate victims in remote regions and discover missing individuals across extensive terrain. The algorithm was primarily trained with individuals of all ages and diverse clothing types, including camouflage, hiking attire, and casual wear. The algorithm aids SAR teams in focusing their efforts on prospective target areas recognised by the system as having a high

probability of containing victims. The ongoing advancement of this technology focuses on real-time regional identification to enhance search efficiency. [b2]

In the context of this SAR framework, this method can improve the identification of individuals beneath rubble and support autonomous detection. To be implemented, the algorithm should be trained across different categories, including dust-covered victims of diverse ages and in different attire, and various bodily regions. This integration can enhance SAR teams' efficiency in identifying victims and effectively performing rescue operations.

The use of two microphones:

This technology was developed to allow a robot to do designated tasks in response to auditory signals. The sound-based controller functions using two microphones that direct the robot toward the sound source. This method contrasts with localization since it focuses on balancing the noise between the two microphones to determine the direction of the sound. This balance of noise is achieved by synchronizing the arrival time discrepancies among the microphones. The benefit is that the mathematical operations render any interference from echoes minimally influential on the robot's overall heading direction. Within the framework of SAR, this technology can augment the robot's capacity to independently align itself with any source of noise. Thus, it aids in locating the victim. [b1]

III. Q1– BASIC PDDL MODEL

This was divided into two subcategories:

- 1) one instance with known location
- 2) second instance requiring exploration

A. Instance 1

Since the location of the victim is known, no explicit exploration is required for each room. Instead, the robot simply navigates to the known room. The robot completes an explicit detection in the room where the victim is located. It then performs the rescue operation, which involves supplying oxygen since the location of the victim is known.

Therefore, the **problem** file is limited to the connected rooms and the anticipated presence of a victim in the dedicated room. The **domain** file involves the robot navigating from room to room until it reaches the room where the victim is located, at which point it completes the scan and provides oxygen. The scanning procedure ensures the presence of the victim. The victim is verified with a high decibel level, a red thermal reading, and a high visual probability.



Fig. 1: Plan Executed for Instance 1

As depicted in Figure 1, the plan aligns with expectations. In comparison, Figure 2 illustrates that the plan adheres to



Fig. 2: Architectural Design of Instance 1

the design selection of *Instance 1*. It explores each room, and when reaching the expected location of the victim, it performs the scanning procedure and initiates the rescue operations of supplying oxygen.

B. Instance 2

Since it requires explicit exploration, the domain and problem have to adjust accordingly.

Since the location is not known, the robot has to go into each room and perform the respective *sensing actions* before it can move on to the next room. The sensing actions are the technology currently used by SAR teams.

Therefore, the **problem** file consists of room connections, the decibel readings, thermal imaging readings, the probability of human detection using the cameras, chemical readings, and battery health.

While the **domain** focuses on scanning every room before moving on to the next, it also involves locating the victim and performing the rescue action. A high decibel level, a red thermal reading, and a high visual probability indicate the presence of a victim. The assessment of the victim's vitals indicates a high CO_2 level for living victims and a high ammonia level for deceased victims.

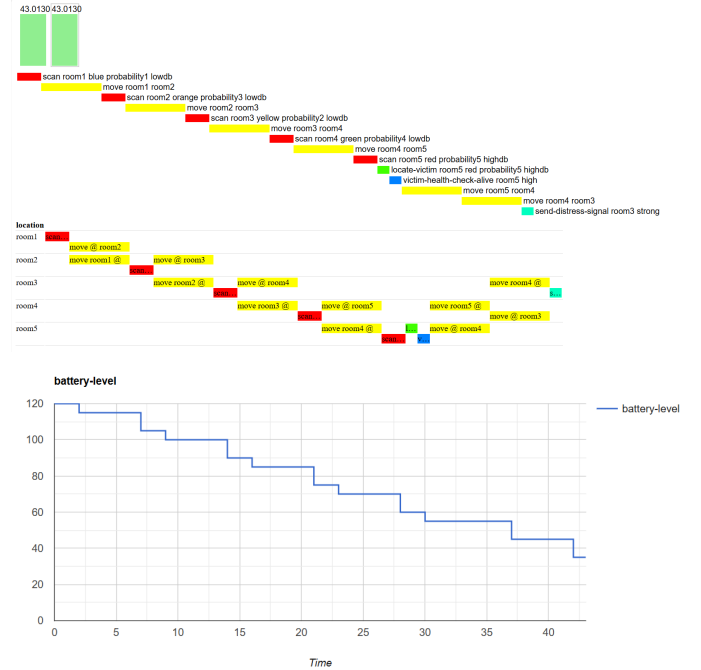


Fig. 3: Plan Executed for Instance 2

These results, highlighted in Figure 3 and Figure 4, show that the planner's expected outcome adheres to the design

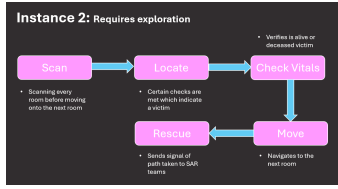


Fig. 4: Architectural Design for Instance 2

selection for *Instance 2*. First, the planner performs an explicit exploration, known as the "scan" durative action, before it can navigate to other rooms. The system includes an explicit "locate" function that deduces a victim's presence in a location, which is indicated by a high thermal reading, a high probability of detection by the camera, and a high decibel reading. From the information gathered from the chemical sensing, it will conclude whether the victim is alive or deceased. The information for each room is part of the **problem** file, as PDDL works with closed-world assumptions.

The explicit "rescue" action is initiated after the victim has been found. The explicit rescue is seen when the robot "sends a distress signal." To complete the task, the robot has to be in a room with a strong connection to send the path it took in finding the victim. If the robot ever loses signal during navigation, it must first locate the victim and then return to a state where a signal was found so it can send the coordinates of where the victim is to SAR teams.

The results also show the battery health; it decreases accordingly with each move and scan.

IV. Q2- PDDL+ MODEL

This consisted of the following:

- Introduce a process modelling victim health degradation.
- Introduce an event representing discovery or failure conditions.
- Show how exploration delays affect rescue success.

The problem file is designed with similar logic to the *second instance*, with additional features such as obstacle detection, victim health, rescue time, and tokens that initiate the move and scan phases.

With the workings of PDDL+, we can include processes, events, and actions. The processes introduced are scanning, moving, victim's health, battery health, and rescue time. The reason for this inclusion is because *scanning* and *moving* take a set amount of time and need to be accounted for as a continuous process where the battery decreases as these actions are computed. The victim's health, battery health, and rescue time constraints are continuous factors that deplete as the rescue and exploration missions are carried out.

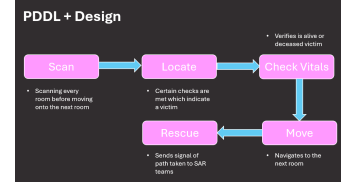
As previously mentioned, these processes correlate to real-life conditions where search and research teams need to monitor whether the rescue mission is feasible, as any unnecessary time spent is a waste of resources during critical situations.

Success is when the victim is found, and the path taken by the robot is sent to SAR teams. Failure occurs when the

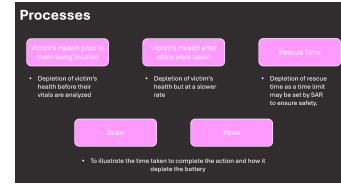
battery dies, the victim isn't found, vital signs drop before data is sent, an obstacle blocks navigation, or the rescue time runs out before data is transmitted. These failures to produce a plan are expected because of the rigid constraints associated with the goal state.

```
Found Plan:
0: (start-scan room1 blue probability1 lowdB)
0: ----waiting---- [5.0]
5.0: (end-scan room1)
5.0: (start-move room1 room2)
5.0: ----waiting---- [15.0]
15.0: (end-move room2)
15.0: (start-scan room2 orange probability3 lowdB)
15.0: ----waiting---- [20.0]
20.0: (end-scan room2)
20.0: (start-move room2 room3)
20.0: ----waiting---- [30.0]
30.0: (end-move room3)
30.0: (start-scan room3 yellow probability3 lowdB)
30.0: ----waiting---- [35.0]
35.0: (end-scan room3)
35.0: (start-move room3 room4)
35.0: ----waiting---- [45.0]
45.0: (end-move room4)
45.0: (start-scan room4 green probability4 lowdB)
45.0: ----waiting---- [50.0]
50.0: (end-scan room4)
50.0: (start-move room4 room5)
50.0: ----waiting---- [60.0]
60.0: (end-move room5)
60.0: (start-scan room5 red probability5 highdB)
60.0: ----waiting---- [65.0]
65.0: (end-scan room5)
65.0: (locate-victim room5)
65.0: (victim-health-check-alive room5)
65.0: (start-move room5 room4)
65.0: ----waiting---- [75.0]
75.0: (end-move room4)
75.0: (bypass-scan)
75.0: (start-move room4 room3)
75.0: ----waiting---- [85.0]
85.0: (end-move room3)
85.0: (send-distress-signal room3)
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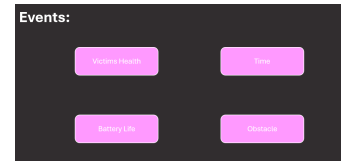
Fig. 5: Executed Plan for PDDL+ scenario



(a) Architectural Design Logic for PDDL+



(b) Designed Processes



(c) Designed Events

Fig. 6: Architectural Layout for PDDL+

As illustrated in Figure 5, the results are as expected; the rooms are scanned before the robot navigates to the next room. The robot will identify the presence of the victim and assess its vitals. Afterwards, it will initiate the rescuing operations, which entails sending a distress signal. Similar to the second

instance, the robot must be in a room with a strong signal to send a distress signal, and it successfully accomplishes this task.

The architectural design for PDDL+ incorporates processes, events, and actions. Figure 6a illustrates the comprehensive design, which follows the same logical sequence as in *Instance 2*. However, the foundational model follows an alternative strategy that shifts from discrete intervals to continuous state evaluation. The specific processes and events are illustrated in Figure 6b and Figure 6c. The integration of these continuous dynamics facilitates a more advanced model that accurately reflects the physical limitations of actual SAR operations.

A. Limitations

A significant limitation of PDDL/PDDL+ is that it relies on topological representation rather than geometric representation of the world. This is why the problem files denote the environment with defined world variables; however, they do not account for geometric conditions. For instance, PDDL/+ cannot model the robot's exact location within the world. This limitation is better illustrated by the scenario in which, after locating a victim, the robot must move closer to analyse the victim's vitals; this allows the chemical sensor system to determine whether the victim is alive or dead. It can not model the robot movements within the room, nor the proximity to the victim.

PDDL/+ functions on closed-world assumptions and operates deterministically. The given world scenario is assumed to be optimal, requiring the fulfillment of all stated activities to accomplish the desired goal of "sending all data." Consequently, if the sensor fails to identify a victim through camera detection due to insufficient visibility, false negatives, or the absence of sound from the victim, the planner cannot handle these uncertainties. Thus, requiring a more heavily modified framework. This modification requires a higher computational complexity, which can cause the planner to malfunction and enter into an infinite evaluation cycle.

A further limitation of PDDL/+ is the rigidity of goal states. An example is if a "fallback" goal were established. This would involve transmitting the data even if the victim's health deteriorates after monitoring their vitals. This scenario creates a disjunctive goal, and the planner would have the choice of which one to achieve. When tested, it revealed that the heuristic planner chooses the path with the least resistance, which is to refrain from moving the robot to allow the victim's health to deteriorate naturally, thereby easily achieving the second goal.

This behavior completely disobeys the logical sequence of the SAR tactic. The ideal logical sequence consists of first attempting to send the goal while the victim is still alive, then acknowledging that the first goal is unattainable, and finally resorting to the second goal. The second goal would still explicitly explore, detect, and rescue.

To prevent this entirely and ensure a stricter goal state, the use of *abort-mission* was implemented as an effect for each

of the events presented in Figure 6c. Therefore, the processes shown in Figure 6b could not continue if the *abort-mission* became true, thereby representing a failure to produce a plan.

A difference highlighted between PDDL and PDDL+ is that, for the *second instance*, all actions had to be converted into durative actions. If not, the system would duplicate the depletion of the battery; the issue was due to a time conflict created by the planner. For example, the battery must be above 20 at the start of any action. However, since durative actions are used, the effect of depleting the battery does not affect the commencement of the action itself. This is the difference between discrete intervals and continuous state evaluation. As demonstrated, the issue is prevented in PDDL+, where it was able to avoid this scenario due to PDDL+ capabilities.

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